

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

AP & AM BASED

1. If $A_1, A_2, A_3, \dots, A_{51}$ are arithmetic means inserted between the numbers a and b , then find the value of $\left(\frac{b+A_{51}}{b-A_{51}}\right) - \left(\frac{A_1+a}{A_1-a}\right)$.

यदि a व b के मध्य $A_1, A_2, A_3, \dots, A_{51}$ है तो $\left(\frac{b+A_{51}}{b-A_{51}}\right) - \left(\frac{A_1+a}{A_1-a}\right)$ का मान बताइये।

Ans. 102

Sol. $a, A_1, A_2, \dots, A_{51}, b$ are in A.P.

$$b = a + 52d$$

$$\left(\frac{b+A_{51}}{b-A_{51}}\right) - \left(\frac{A_1+a}{A_1-a}\right) = \left(\frac{b+a+51d}{b-a-51d}\right) - \left(\frac{2a+d}{d}\right)$$

$$= \left(\frac{a+52d+a+51d}{d}\right) - \left(\frac{2a+d}{d}\right)$$

$$= \frac{2a+103d-2a-d}{d}$$

$$= 102$$

2. Let $a_1, a_2, a_3, \dots, a_{11}$ be real numbers satisfying $a_1 = 15, 27 - 2a_2 > 0$ and $a_k = 2a_{k-1} - a_{k-2}$ for $k = 3, 4, \dots, 11$. If

$$\frac{a_1^2 + a_2^2 + \dots + a_{11}^2}{11} = 90, \text{ then the value of } \frac{a_1 + a_2 + \dots + a_{11}}{11} \text{ is equal to.} \quad [\text{JEE-2010}]$$

यदि $a_1, a_2, a_3, \dots, a_{11}$ वास्तविक संख्याएँ हैं तथा $a_1 = 15, 27 - 2a_2 > 0$ और $a_k = 2a_{k-1} - a_{k-2}$ जहाँ $k = 3, 4, \dots, 11$. यदि

$$\frac{a_1^2 + a_2^2 + \dots + a_{11}^2}{11} = 90 \text{ तो } \frac{a_1 + a_2 + \dots + a_{11}}{11} \text{ का मान बताइये।} \quad [\text{JEE-2010}]$$

Ans. 0

Sol. $a_1 = 15$

$$\frac{a_k + a_{k-2}}{2} = a_{k-1} \text{ for } k = 3, 4, \dots, 11 \text{ के लिए} \Rightarrow a_1, a_2, \dots, a_{11} \text{ are in AP समान्तर श्रेणी में हैं।}$$

$$a_1 = a = 15$$

$$\frac{a_1^2 + a_2^2 + \dots + a_n^2}{11} = 90 \Rightarrow \frac{(15)^2 + (15+d)^2 + \dots + (15+10d)^2}{11} = 90$$

$$\Rightarrow 7d^2 + 30d + 27 = 0 \Rightarrow d = -3 \text{ or } -\frac{9}{7}$$

$$\text{Since चूंकि } 27 - 2a_2 > 0 \Rightarrow a_2 < \frac{27}{2} \Rightarrow d = -3$$

$$\frac{a_1 + a_2 + \dots + a_{11}}{11} = \frac{11}{2} \frac{[30 + 10(-3)]}{11} = 0$$

3. Let $a_1, a_2, a_3, \dots, a_{100}$ be an arithmetic progression with $a_1 = 3$ and $S_p = \sum_{i=1}^p a_i$, $1 \leq p \leq 100$. For any integer n

with $1 \leq n \leq 20$, let $m = 5n$. If $\frac{S_m}{S_n}$ does not depend on n , then a_2 is. [JEE-2011]

माना $a_1, a_2, a_3, \dots, a_{100}$ एक समान्तर श्रेणी में हो जिसके लिये $a_1 = 3$ तथा $S_p = \sum_{i=1}^p a_i$, $1 \leq p \leq 100$, किसी पूर्णांक n के लिये

$1 \leq n \leq 20$. यदि $m = 5n$ तथा $\frac{S_m}{S_n}$, n के मान पर निर्भर नहीं करता है तो a_2 होगा। [JEE-2011]

Ans. 3 or 9

Sol. $\frac{S_m}{S_n} = \frac{S_{5n}}{S_n} = \frac{\frac{5n}{2} [6 + (5n-1)d]}{\frac{n}{2} [6 + (n-1)d]} = \frac{5[(6-d) + 5nd]}{[(6-d) + nd]}$; $d = 6$ or $d = 0$

Now if $d = 0$ then $a_2 = 3$ else $a_2 = 9$ for single choice more appropriate choice is 9, but in principal, question seems to have an error.

$$\therefore a_2 = 3 + 6 = 9$$

GP & GM BASED

4. The 1st, 2nd and 3rd terms of an arithmetic series are a, b and a^2 where ' a ' is negative. The 1st, 2nd and 3rd terms of a geometric series are a, a^2 and b find the

- value of a and b
- sum of infinite geometric series if it exists. If no then find the sum to n terms of the G.P.
- sum of the 40 term of the arithmetic series.

एक समानान्तर श्रेणी के प्रथम, द्वितीय तथा तृतीय पद क्रमशः a, b तथा a^2 है जहाँ ' a ' एक ऋणात्मक संख्या है। एक गुणोत्तर श्रेणी के प्रथम तीन पद a, a^2 तथा b है तो

- a एवं b का मान निकालिये।

- (b) अनन्त गुणोत्तर श्रेणी का योग निकालिये (यदि अस्तित्व में हो) यदि नहीं तो G.P के n पदों का योग निकालिये।
 (c) समान्तर श्रेणी के 40 पदों का योग निकालिये।

Ans. (a) $a = -\frac{1}{2}$, $b = -\frac{1}{8}$; (b) $-\frac{1}{3}$; (c) $\frac{545}{2}$

Sol. (a) a, b, a^2 are in A.P. ($a < 0$)
 a, a^2, b are in G.P.
 $\Rightarrow 2b = a + a^2$
 $(a^2)^2 = ab$

$$a^4 = a \left(\frac{a + a^2}{2} \right)$$

$$a^2 \left[a^2 - \left(\frac{a+1}{2} \right) \right] = 0$$

$$a^2 = 0 \Rightarrow \text{Not possible}$$

$$a^2 - \left(\frac{a+1}{2} \right) = 0$$

$$2a^2 - a - 1 = 0$$

$$2a^2 - 2a + a - 1 = 0$$

$$2a(a-1) + (a-1) = 0$$

$$a = -\frac{1}{2}, a = 1 \text{ (reject)}$$

$$a = -\frac{1}{2}$$

$$2b = a + a^2 = \frac{-1}{2} + \frac{1}{4} = \frac{-1}{4}$$

$$b = \frac{-1}{8}$$

$$(b) \text{ Sum of infinite G.P.} = \frac{a}{1 - \left(\frac{a^2}{a} \right)} = \frac{a}{1-a}$$

$$= \frac{\left(-\frac{1}{2} \right)}{1 + \frac{1}{2}} = -\frac{1}{3}$$

$$(c) \text{ Sum of 40 terms of A.P.}$$

$$\begin{aligned}
S &= \frac{40}{2} \left[2 \left(-\frac{1}{2} \right) + 39 \left(-\frac{1}{8} + \frac{1}{2} \right) \right] \\
&= 20 \left[-1 + 39 \left(\frac{-1+4}{8} \right) \right] \\
&= 20 \left[\frac{-8+39 \times 3}{8} \right] \\
&= \frac{545}{2}
\end{aligned}$$

5. For $n = 1, 2, 3, \dots$, let $A_n = \left(\frac{3}{4}\right) - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + \dots + (-1)^{n-1} \left(\frac{3}{4}\right)^n$, and $B_n = 1 - A_n$,

Find the smallest natural number n_0 such that $B_n > A_n$ for all $n \geq n_0$.

[JEE-2006]

यदि $A_n = \left(\frac{3}{4}\right) - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + \dots + (-1)^{n-1} \left(\frac{3}{4}\right)^n$ तथा $B_n = 1 - A_n$ है तो वो प्राकृतिक संख्या n_0 बताइए जिसके लिये

$B_n > A_n, \forall n > n_0$.

[JEE-2006]

Ans. $n_0 = 6$

Sol. $a_n = \frac{3}{4} - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + \dots + (-1)^{n-1} \left(\frac{3}{4}\right)^n, b_n = 1 - a_n = \frac{3}{4} \left[\frac{1 - \left(-\frac{3}{4}\right)^n}{\left(1 + \frac{3}{4}\right)} \right] = \frac{3}{7} \left[1 - \left(-\frac{3}{4}\right)^n \right]$

$$b_n > a_n \Rightarrow 1 - a_n > a_n$$

$$2a_n < 1 \Rightarrow \frac{6}{7} - \frac{6}{7} \left(-\frac{3}{4}\right)^n < 1$$

$$-\frac{6}{7} \left(-\frac{3}{4}\right)^n < \frac{1}{7} \Rightarrow -3(-3)^n < 2^{2n-1} \Rightarrow (-1)^{n+1} 3^{n+1} < 2^{2n-1}$$

For n even inequality is always valid

For n , odd ; $n+1 \rightarrow$ even

$3^{n+1} < 2^{2n-1}$; $n=1, 3^2 < 2^1$, (in valid)

$n=3, 3^4 < 2^5, 81 < 32$ (in valid) ; $n=5, 3^6 < 2^9$ (in valid)

$n=7, 3^8 < 2^{13}, 6561 < 8192$ (valid) ; min. natural number $n_0=6$

6. If $x = 1 + a + a^2 + a^3 + \dots$ to ∞ ($|a| < 1$) and $y = 1 + b + b^2 + b^3 + \dots$ to ∞ ($|b| < 1$), then prove that

$$1 + ab + a^2b^2 + a^3b^3 + \dots \infty = \frac{xy}{x+y-1}.$$

यदि $x = 1 + a + a^2 + a^3 + \dots$ से ∞ ($|a| < 1$) तक तथा $y = 1 + b + b^2 + b^3 + \dots$ से ∞ ($|b| < 1$) तक है, तो सिद्ध कीजिए कि

$$1 + ab + a^2b^2 + a^3b^3 + \dots \text{ से } \infty \text{ तक} = \frac{xy}{x+y-1} \text{ है।}$$

Ans.

Sol. $x = 1 + a + a^2 + \dots = \frac{1}{1-a}$

$$y = 1 + b + b^2 + \dots = \frac{1}{1-b}$$

$$1 + ab + a^2b^2 + \dots = \frac{1}{1-ab}$$

$$\frac{xy}{x+y-1} = \frac{\left(\frac{1}{1-a}\right)\left(\frac{1}{1-b}\right)}{\left(\frac{1}{1-a}\right) + \left(\frac{1}{1-b}\right) - 1} = \frac{1}{(1-b) + (1-a) - (1-a)(1-b)}$$

$$= \frac{1}{2-a-b-(1-a-b+ab)}$$

$$= \frac{1}{1-ab} \quad \text{H.P.}$$

7. There are 25 trees at equal distances of 5 m in a line with a well, the distance of the well from the nearest tree being 10 m. A gardener waters all the trees separately starting from the well and he returns to the well after watering each tree to get water for the next. Find the total distance, the gardener will cover in order to water all the trees.

यदि किसी रेखा में 25 वृक्ष एक दूसरे से 5 m की दूरी पर हैं तथा इसी रेखा में एक कुआँ है जो सबसे नजदीकी पेड़ से 10 m दूर है। एक बागवान कुएँ से एक वृक्ष को पानी देकर, दूसरे पेड़ के लिये पानी लाने के लिए कुएँ पर जाता है तथा इसी प्रकार प्रत्येक वृक्ष को पानी देता है। तो वह कुल दूरी बताओं जो बागवान सभी पेड़ों को पानी देने में तय करता है।

Ans. 3500

Sol. $T_{25} = 10 + 24 \times 5 = 130$

$$\text{Total distance} = 2 \times \frac{25}{2} [2 \times 10 + (24)5]$$

$$= 25 [20 + 120]$$

$$= 25 [140]$$

$$= 3500$$

8. The sum of infinite number of terms of a G.P. is 4 and the sum of their cubes is 192. Find the series.

एक अन्तत पदो वाली गु. श्रे. के पदो का योग 4 है। और उनके घनो का योग 192 है। श्रेणी ज्ञात कीजिए।

Ans. 6, -3, 3/2,

Sol. $\frac{a}{1-r} = 4 \Rightarrow \frac{a^3}{1-r^3} = 192 \Rightarrow \frac{(1-r)^3}{1-r^3} = \frac{192}{(4)^3} \Rightarrow \frac{(1-r)^2}{1+r+r^2} = 3$

$$\Rightarrow 1+r^2-2r=3+3r+3r^2 \Rightarrow 2r^2+5r+2=0 \Rightarrow (2r+1)(r+2)=0$$

$$r=-1/2, r=-2(\text{rejected}) \quad \text{When } r=-1/2, a=6 \quad \text{so series is } 6, -3, 3/2, \dots$$

Hindi. $\frac{a}{1-r} = 4 \Rightarrow \frac{a^3}{1-r^3} = 192 \Rightarrow \frac{(1-r)^3}{1-r^3} = \frac{192}{(4)^3} \Rightarrow \frac{(1-r)^2}{1+r+r^2} = 3$

$$\Rightarrow 1+r^2-2r=3+3r+3r^2 \Rightarrow 2r^2+5r+2=0 \Rightarrow (2r+1)(r+2)=0$$

$$r=-1/2, r=-2 (\text{अस्वीकार्य}) \quad \text{जब } r=-1/2, a=6 \quad \text{अतः श्रेणी } 6, -3, 3/2, \dots \text{ है।}$$

AM GM HM

9. The harmonic mean of two numbers is 4. The airthmetic mean A & the geometric mean G satisfy the relation $2A + G^2 = 27$. Find the two numbers.

दो संख्याओं का हरात्मक माध्य 4 है। उनका समान्तर माध्य 'A' एवं गुणोत्तर माध्य 'G' है जो सम्बन्ध $2A + G^2 = 27$ को संतुष्ट करते हैं। दोनों संख्याएँ निकालिए।

Ans. 6, 3

Sol. $H = 4 = \frac{2ab}{a+b} \Rightarrow a+b = \frac{ab}{2}$

$$A = \frac{a+b}{2}, \quad G = \sqrt{ab}$$

$$2A + G^2 = a+b+ab = 27$$

$$\frac{ab}{2} + ab = 27$$

$$\frac{3ab}{2} = 27$$

$$ab = 18 \Rightarrow a+b = 9$$

$$x^2 - (a+b)x + ab = 0$$

$$x^2 - 9x + 18 = 0$$

$$(x-6)(x-3) = 0$$

$$x = 6, 3$$

10. The AM of two numbers exceeds their GM by 15 & HM by 27. Find the numbers.

दो संख्याओं का स.मा. उनके गु. मा. से 15 बढ़ा है एवं ह.मा. से 27 बढ़ा है। दोनों संख्याएँ निकालिए।

Ans. 120, 30

Sol. $\frac{a+b}{2} = \sqrt{ab} + 15, \quad \frac{a+b}{2} = \frac{2ab}{a+b} + 27$

$$2\sqrt{ab} = (a+b) - 30, \quad (a+b)^2 = (2\sqrt{ab})^2 + 54(a+b)$$

$$(a+b)^2 = [(a+b) - 30]^2 + 54(a+b)$$

$$\text{Let } a+b = x$$

$$x^2 = x^2 + 900 - 60x + 54x$$

$$6x = 150$$

$$x = 150$$

$$a+b=150 \Rightarrow \frac{150}{2} \cdot 15 = \sqrt{ab}$$

$$60 = \sqrt{ab}$$

$$ab = 3600$$

$$y^2 + \{-(a+b)\}y + ab = 0$$

$$y^2 - 150y + 3600 = 0$$

$$(y-120)(y-30) = 0$$

$$y = 30, 120$$

11. If a, b, c are positive real numbers, then prove that $[(1+a)(1+b)(1+c)]^7 > 7^7 a^4 b^4 c^4$. **[JEE-2004]**

यदि a, b, c धनात्मक वास्तविक संख्याएँ हैं तो सिद्ध कीजिये। $[(1+a)(1+b)(1+c)]^7 > 7^7 a^4 b^4 c^4$ **[JEE-2004]**

Ans.

Sol. $(1+a)(1+b)(1+c) = (1+a+b+ab)(1+c)$

$$= 1 + a + b + ab + c + ac + bc + abc$$

$$= 1 + a + b + c + ab + bc + ca + abc$$

$$\text{A.M.} \geq \text{G.M.}$$

$$\frac{a+b+c+ab+bc+ca+abc}{7} \geq [a^4 b^4 c^4]^{\frac{1}{7}}$$

$$(a+b+c+ab+bc+ca+abc) \geq 7(a^4 b^4 c^4)^{\frac{1}{7}}$$

$$1+a+b+c+ab+bc+ca+abc \geq 7(a^4 b^4 c^4)^{\frac{1}{7}}$$

$$[(1+a)(1+b)(1+c)]^7 \geq 7^7(a^4 b^4 c^4)$$

12. The minimum value of the sum of real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ and a^{10} with $a > 0$ is.

[JEE-2011]

वास्तविक संख्याएँ $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ तथा a^{10} जहाँ $a > 0$ का योग का न्यूनतम मान होगा।

[JEE-2011]

Ans. 8

Sol. A.M. $\geq$ G.M.

$$\frac{\frac{1}{a^5} + \frac{1}{a^4} + \frac{1}{a^3} + \frac{1}{a^3} + \frac{1}{a^3} + 1 + a^8 + a^{10}}{8} \geq \left(\frac{1}{a^5} \cdot \frac{1}{a^4} \cdot \frac{1}{a^3} \cdot \frac{1}{a^3} \cdot \frac{1}{a^3} \cdot 1 \cdot a^8 \cdot a^{10} \right)^{1/8}$$

$$\Rightarrow \frac{1}{a^5} + \frac{1}{a^4} + \frac{3}{a^3} + 1 + a^8 + a^{10} \geq 8(1)^{1/8}$$

$$\Rightarrow \text{minimum value of } \frac{1}{a^5} + \frac{1}{a^4} + \frac{3}{a^3} + 1 + a^8 + a^{10} = 8$$

Hindi A.M. $\geq$ G.M.

$$\frac{\frac{1}{a^5} + \frac{1}{a^4} + \frac{1}{a^3} + \frac{1}{a^3} + \frac{1}{a^3} + 1 + a^8 + a^{10}}{8} \geq \left(\frac{1}{a^5} \cdot \frac{1}{a^4} \cdot \frac{1}{a^3} \cdot \frac{1}{a^3} \cdot \frac{1}{a^3} \cdot 1 \cdot a^8 \cdot a^{10} \right)^{1/8}$$

$$\Rightarrow \frac{1}{a^5} + \frac{1}{a^4} + \frac{3}{a^3} + 1 + a^8 + a^{10} \geq 8(1)^{1/8}$$

$$\Rightarrow \frac{1}{a^5} + \frac{1}{a^4} + \frac{3}{a^3} + 1 + a^8 + a^{10} \text{ का } a = 1 \text{ पर न्यूनतम मान } = 8$$

HP & HM BASED

13. If the 10th term of an HP is 21 and 21st term of the same HP is 10, then find the 210th term.

यदि एक HP का 10वाँ पद 21 है एवं 21वाँ पद 10 है तो 210वाँ पद बताइये।

Ans. 1

Sol. $T_{10} = 21, T_{21} = 10$

$$t_{10} = \frac{1}{21}, t_{21} = \frac{1}{10}$$

$$\frac{1}{a} + 9d = \frac{1}{21} \quad \dots(1)$$

$$\frac{1}{a} + 20d = \frac{1}{10} \quad \dots(2)$$

$$(1) - (2)$$

$$-11d = \frac{1}{21} - \frac{1}{10}$$

$$-11d = \frac{10-21}{210}$$

$$d = \frac{1}{210} \Rightarrow \frac{1}{a} = \frac{1}{210}$$

$$t_{210} = \frac{1}{210} + (210-1) \times \frac{1}{210} = 1$$

$$T_{210} = 1$$

14. The p^{th} term T_p of H.P. is $q(p+q)$ and q^{th} term T_q is $p(p+q)$ when $p > 2$, $q > 2$. Prove that

एक HP का $p^{\text{वाँ}}$ पद $T_p = q(p+q)$ है तथा $q^{\text{वाँ}}$ पद $T_q = p(p+q)$ है जब $p > 2$, $q > 2$ है। सिद्ध कीजिये।

(a) $T_{p+q} = pq$; (b) $T_{pq} = p+q$; (c) $T_{p+q} > T_{pq}$

Ans.

Sol. $T_p = q(p+q) \Rightarrow t_p = \frac{1}{q(p+q)} = \frac{1}{a} + (p-1)d \quad \dots(1)$

$$T_q = p(p+q) \Rightarrow t_q = \frac{1}{p(p+q)} = \frac{1}{a} + (q-1)d \quad \dots(2)$$

$$(1) - (2) : \left(\frac{1}{p+q} \right) \left(\frac{1}{q} - \frac{1}{p} \right) = (p-q)d$$

$$\frac{1}{(p+q)pq} = d = \frac{1}{a}$$

$$(a) \quad t_{p+q} = \frac{1}{(p+q)pq} + (p+q-1) \frac{1}{(p+q)(pq)}$$

$$t_{p+q} = \frac{1}{pq}$$

$$\Rightarrow T_{p+q} = pq$$

$$(b) \quad t_{pq} = \frac{1+(pq-1)}{pq(p+q)} = \frac{1}{p+q}$$

$$T_{pq} = p+q$$

$$(c) \quad T_{pq} = pq > p+q \quad (\because p > 2, q > 2)$$

AGP

15. Let ' σ ' denotes the sum of the infinite series $\sum_{n=1}^{\infty} \left(\frac{n^2 + 2n + 3}{2^n} \right)$. Compute the value of $(1^3 + 2^3 + 3^3 + \dots + \sigma^3)$.

माना ' σ ' अनन्त श्रेणी $\sum_{n=1}^{\infty} \left(\frac{n^2 + 2n + 3}{2^n} \right)$ का योग है। तो $(1^3 + 2^3 + 3^3 + \dots + \sigma^3)$ के मान की गणना कीजिये।

Ans. 8281

Sol. $\sigma = \sum_{n=1}^{\infty} \left(\frac{n^2 + 2n + 3}{2^n} \right)$

$$\sigma = \frac{6}{2} + \frac{11}{2^2} + \frac{18}{2^3} + \frac{27}{2^4} + \dots \quad \dots(1)$$

$$\frac{\sigma}{2} = 0 + \frac{6}{2^2} + \frac{11}{2^3} + \frac{18}{2^4} + \dots \quad \dots(2)$$

$$(1) - (2)$$

$$\frac{\sigma}{2} = 3 + \frac{5}{2^2} + \frac{7}{2^3} + \frac{9}{2^4} + \dots \quad \dots(3)$$

$$\frac{\sigma}{4} = 0 + \frac{3}{2} + \frac{5}{2^3} + \frac{7}{2^4} + \dots \quad \dots(4)$$

$$(3) - (4)$$

$$\frac{\sigma}{4} = \left(3 + \frac{5}{4} - \frac{3}{2} \right) + \left(\frac{2}{2^3} + \frac{2}{2^4} + \dots \right)$$

$$= \frac{11}{4} + \frac{1}{2^2} \left(1 + \frac{1}{2} + \frac{1}{2^2} + \dots \right)$$

$$\frac{\sigma}{4} = \frac{11}{4} + \frac{1}{4} \left(\frac{1}{1 - \frac{1}{2}} \right)$$

$$\sigma = 11 + 2$$

$$\sigma = 13$$

$$\text{Now } 1^3 + 2^3 + 3^3 + \dots + 13^3 = \left(\frac{13 \times 14}{2} \right)^2$$

$$= (91)^2$$

$$= 8281$$

16. If total number of runs scored in n matches is $\left(\frac{n+1}{4}\right)(2^{n+1} - n - 2)$ where $n > 1$, and the runs scored in the k^{th} match are given by $k \cdot 2^{n+1-k}$, where $1 \leq k \leq n$. Find n . [JEE-2005]

यदि n मैचों में अर्जित कुल रनों की संख्या $\left(\frac{n+1}{4}\right)(2^{n+1} - n - 2)$ जहाँ $n > 1$ है एवं k वें मैच में अर्जित रनों को यदि $k \cdot 2^{n+1-k}$ से निरूपित किया जाता है जहाँ $1 \leq k \leq n$ है। n का मान निकालिए। [JEE-2005]

Ans. $n = 7$

Sol. Total run in n matches $= \left(\frac{n+1}{4}\right)(2^{n+1} - n - 2)$

$$T_k = k \cdot 2^{n+1-k}$$

$$S_n = \sum_{k=1}^n \left(2^{n+1}\right) \left(\frac{k}{2^k}\right)$$

$$S_n = 2^{n+1} \left[\frac{1}{2} + \frac{1}{2^2} + \frac{3}{2^3} + \frac{4}{2^4} + \dots + \frac{n}{2^n} \right]$$

$$S = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \frac{4}{2^4} + \dots + \frac{n}{2^n} \quad \dots(1)$$

$$\frac{S}{2} = 0 + \frac{1}{2^2} + \frac{2}{2^3} + \frac{3}{2^4} + \dots + \frac{n-1}{2^n} + \frac{n}{2^{n+1}} \quad \dots(2)$$

$$(1) - (2)$$

$$\frac{S}{2} = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \dots + \frac{1}{2^n} - \frac{n}{2^{n+1}}$$

$$S = \left(1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}}\right) \frac{n}{2^n}$$

$$S = \frac{1 \left(\frac{1}{2}\right)^n - 1}{\frac{1}{2} - 1} - \frac{n}{2^n}$$

$$S = \frac{1 - 2^n}{2^n \left(-\frac{1}{2}\right)} - \frac{n}{2^n}$$

$$S = \frac{2^n - 1}{2^{n-1}} - \frac{n}{2^n}$$

$$S = \frac{2^{n+1} - 2 - n}{2^n}$$

$$S_n = 2^{n+1} \left(\frac{2^{n+1} - n - 2}{2^n} \right) = 2(2^{n+1} - n - 2)$$

$$= \left(\frac{n+1}{4} \right) (2^{n+1} - n - 2)$$

$$\Rightarrow 2 = \frac{n+1}{4} \Rightarrow n = 7$$

MISCELLANEOUS SEQUENCES

17. If $S_1, S_2, S_3, \dots, S_n, \dots$ are the sums of infinite geometric series whose first terms are $1, 2, 3, \dots, n, \dots$ and whose common ratios are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n+1}, \dots$ respectively, then find the value of $\sum_{r=1}^{2n-1} S_r^2$.

यदि $S_1, S_2, S_3, \dots, S_n, \dots$ अनन्त गुणोत्तर श्रेणी के योग हैं जिनके प्रथम पद $1, 2, 3, \dots, n, \dots$ एवं सर्वानुपात

$\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n+1}, \dots$ क्रमशः हैं तो $\sum_{r=1}^{2n-1} S_r^2$ का मान निकालिए।

Ans. $\frac{n(2n+1)(4n+1)}{3} - 1$

Sol. $S_1 = \frac{1}{1 - \frac{1}{2}}, S_2 = \frac{2}{1 - \frac{1}{3}}, S_3 = \frac{3}{1 - \frac{1}{4}}, \dots$

$$S_1 = \frac{2}{1}, S_2 = \frac{2 \cdot 3}{2}, S_3 = \frac{3 \cdot 4}{3} \dots$$

$$S_1 = 2, S_2 = 3, S_3 = 4 \dots$$

$$\sum_{r=1}^{2n-1} (S_r)^2 = S_1^2 + S_2^2 + S_3^2 + \dots + S_{2n-1}^2$$

$$= (2)^2 + (3)^2 + (4)^2 + \dots + (2n)^2$$

$$= \frac{2n(2n+1)(4n+1)}{6} - 1$$

18. Find the n th term and the sum to n terms of the sequence :

निम्न श्रेणियों का n वाँ पद एवं n पदों का योग बताइए :

(i) $1+5+13+29+61+\dots$ (ii) $6+13+22+33+\dots$

Ans. (i) $2^{n+1}-3$; $2^{n+2}-4-3n$ (ii) n^2+4n+1 ; $(1/6)n(n+1)(2n+13)+n$

Sol. $S = 1 + 5 + 13 + 29 + 61 + \dots T_n$ $S = 1 + 5 + 13 + 29 + \dots T_n$

$$0 = 1 + 4 + 8 + 16 + 32 + \dots T_n \quad T_n = 1 + 4(2^{n-1} - 1) = 2^{n+1} - 3$$

$$S_n = \Sigma(2^{n+1} - 3) = (2^2 + 2^3 + \dots 2^{n+1}) - 3n = 2^2 \left(\frac{2^n - 1}{2 - 1} \right) - 3n = 2^{n+2} - 4 - 3n$$

(b) $S = 6 + 13 + 22 + 33 + \dots T_n$ $S = 6 + 13 + 22 + \dots T_n$
 $T_n = 6 + 7 + 9 + 11 \dots$ $= 1 + 5 + 7 + 9 + 11 \dots$

$$= 1 + \frac{n}{2}(10 + (n-1)2) = n^2 + 4n + 1 \quad S_n = \Sigma T_n = \frac{1}{6}n(n+1)(2n+13) + n$$

19. If the sum $\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}} + \sqrt{1+\frac{1}{2^2}+\frac{1}{3^2}} + \sqrt{1+\frac{1}{3^2}+\frac{1}{4^2}} + \dots + \sqrt{1+\frac{1}{(1999)^2}+\frac{1}{(2000)^2}}$ equal to $n - \frac{1}{n}$ where $n \in \mathbb{N}$. Find n .

यदि योग $\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}} + \sqrt{1+\frac{1}{2^2}+\frac{1}{3^2}} + \sqrt{1+\frac{1}{3^2}+\frac{1}{4^2}} + \dots + \sqrt{1+\frac{1}{(1999)^2}+\frac{1}{(2000)^2}}$ का मान $n - \frac{1}{n}$ जहाँ $n \in \mathbb{N}$

के बराबर है तो n का मान बताइए।

Ans. $n = 2000$

Sol.
$$T_r = \sqrt{1 + \frac{1}{r^2} + \frac{1}{(r+1)^2}}$$

$$= \sqrt{\frac{(r(r+1))^2 + (r+1)^2 + r^2}{(r(r+1))^2}}$$

$$T_r = \frac{r(r+1)+1}{r(r+1)} = 1 + \frac{1}{r(r+1)}$$

$$S = \sum_{r=1}^{1999} T_r = \sum_{r=1}^{1999} \left(1 + \frac{1}{r(r+1)} \right)$$

$$= 1999 + \sum_{r=1}^{1999} \left(\frac{(r+1)-r}{r(r+1)} \right)$$

$$= 1999 + \left(1 - \frac{1}{1999+1}\right)$$

$$n - \frac{1}{n} = 2000 - \frac{1}{2000}$$

$$n = 2000$$

20. The series of natural numbers is divided into groups (1); (2, 3, 4); (5, 6, 7, 8, 9); ... and so on. Show that the sum of the numbers in the n th group is $(n-1)^3 + n^3$.

प्राकृत संख्याओं की श्रेणियों को निम्नलिखित समूह (1); (2, 3, 4); (5, 6, 7, 8, 9); ... में बांटा जाता है, तो सिद्ध कीजिए कि n वें समूह में संख्याओं का योग $(n-1)^3 + n^3$ है।

Ans.

Sol. (1), (2,3,4), (5,6,7,8,9),

Total number of digits in T_n term

$$1, 3, 5, 7, \dots, (1 + (n-1)2) = 2n-1$$

First term of T_n term

$$S = 1 + 2 + 5 + 10 + \dots + T_n \quad \dots(1)$$

$$S = 0 + 1 + 2 + 5 + \dots + T_{n-1} + T_n \quad \dots(2)$$

$$(1) - (2)$$

$$0 = 1 + 1 + 3 + 5 + \dots + T_n$$

(n-1) term

$$T_n = 1 + \left(\frac{n-1}{2}\right)(2 + (n-2) \times 2)$$

$$n = 1 + (n-1)^2$$

n^{th} group

$$(1 + (n-1)^2, 2 + (n-1)^2, \dots, (2n-1) \text{ terms})$$

$$S = \left(\frac{2n-1}{2}\right) \left[2 \left(1 + (n-1)^2\right) + (2n-2) \right]$$

$$= (2n-1) [1 + (n-1)^2 + n-1]$$

$$= (2n-1) [(n-1)^2 + n]$$

$$S = (2n-1) [n^2 + 1 - n]$$

$$= (2n-1) [n^2 - n + 1]$$

$$= 2n^3 - 2n^2 + 2n - n^2 + n - 1$$

$$= 2n^3 - 3n^2 + 3n - 1$$

$$S = (n-1)^3 + n^3$$

21. Sum the following series to n terms and to infinity :

निम्नलिखित श्रेणियों के n पदों एवं अनन्त पदों का योग बताइए :

$$(i) \frac{1}{1.4.7} + \frac{1}{4.7.10} + \frac{1}{7.10.13} + \dots$$

$$(ii) \sum_{r=1}^n r(r+1)(r+2)(r+3)$$

$$(iii) \sum_{r=1}^n \frac{1}{4r^2 - 1}$$

$$(iv) \frac{1}{2.4} + \frac{1.3}{2.4.6} + \frac{1.3.5}{2.4.6.8} + \dots$$

Ans. (i) $s_n = (1/24) - [1/\{6(3n+1)(3n+4)\}]$; $s_\infty = 1/24$ (ii) $(1/5) n(n+1)(n+2)(n+3)(n+4)$

$$(iii) n/(2n+1)$$

$$(iv) S_n = \left[\frac{1}{2} - \frac{1.3.5 \dots (2n-1)(2n+1)}{2.4.6 \dots (2n)(2n+2)} \right]; S_\infty = \frac{1}{2}$$

Sol. (a) $\frac{1}{1.4.7} + \frac{1}{4.7.10} + \frac{1}{7.10.13} + \dots$

$$T_n = \frac{1}{(3n-2)(3n+1)(3n+4)}$$

$$= \frac{1}{3n+1} \left(\frac{1}{3n-2} - \frac{1}{(3n+4)} \right) \frac{1}{6}$$

$$= \left(\frac{1}{(3n-2)(3n+1)} - \frac{1}{(3n+1)(3n+4)} \right) \frac{1}{6}$$

$$S_n = \frac{1}{6} \left[\frac{1}{1.4} - \frac{1}{4.7} + \frac{1}{4.7} - \frac{1}{7.10} + \frac{1}{(3n-2)(3n+1)} - \frac{1}{(3n+1)(3n+4)} \right]$$

$$= \frac{1}{6} \left[\frac{1}{4} - \frac{1}{(3n+1)(3n+4)} \right]$$

$$\text{now } S_\infty = \frac{1}{24}$$

(b) $T_n = n(n+1)(n+2)(n+3)$

$$= \frac{1}{5} [n(n+1)(n+2)(n+3)((n+4) - (n-1))]$$

$$= \frac{1}{5} [n(n+1)(n+2)(n+3)(n+4) - (n-1)(n)(n+2)(n+3)]$$

$$\Sigma T_n = \left(\frac{1}{5}\right)n(n+1)(n+2)(n+3)(n+4)$$

$$(c) T_n = \frac{1}{4n^2 - 1} = \frac{1}{2} \left[\frac{1}{2n-1} - \frac{1}{2n+1} \right]$$

$$S_n = \frac{1}{2} \left[1 - \frac{1}{2n+1} \right]$$

$$S_\infty = \frac{1}{2}$$

$$(d) T_n = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)[(2n+2) - (2n+1)]}{2 \cdot 4 \cdot 6 \dots (2n+2)}$$

$$= \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n} - \frac{1 \cdot 3 \cdot 5 \dots (2n+1)}{2 \cdot 4 \cdot 6 \dots (2n+2)}$$

$$= \left(\frac{1}{2} - \frac{1 \cdot 3}{2 \cdot 4} \right) + \left(\frac{1 \cdot 3}{2 \cdot 4} - \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \right) + \dots + \left(\frac{1 \cdot 3 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n} \right) - \frac{1 \cdot 3 \cdot 5 \dots (2n+1)}{2 \cdot 4 \cdot 6 \dots (2n+2)}$$

$$S_n = \frac{1}{2} - \frac{1 \cdot 3 \cdot 5 \dots (2n+1)}{2 \cdot 4 \cdot 6 \dots (2n+2)},$$

$$S_n = \frac{1}{2} - \frac{1}{2^{n+1}} \left(\frac{1 \cdot 3 \cdot 5 \dots (2n+1)}{1 \cdot 2 \cdot 3 \dots (n+1)} \right)$$

$$S_\infty = \frac{1}{2}$$

22. If $\sum_{r=1}^{20} r^3 = a$, $\sum_{r=1}^{20} r^2 = b$ then sum of products of 1, 2, 3, 4 20 taking two at a time is.

यदि $\sum_{r=1}^{20} r^3 = a$, $\sum_{r=1}^{20} r^2 = b$ है, तो 1, 2, 3, 4 20 को क्रमवार युग्म (दो एक साथ) का गुणनफल होगा।

Ans. $\frac{a-b}{2}$

Sol. $\sum_{r=1}^{20} r^3 = a$, $\sum_{r=1}^{20} r^2 = b$

$$a = 1 + 2^3 + 3^3 + 4^3 + \dots + (20)^3 = \left[\frac{20 \times 21}{2} \right]^2$$

$$b = 1 + 2^2 + 3^2 + 4^2 + \dots + (20)^2 = \left[\frac{20 \times 21 \times 41}{6} \right]$$

$$(1 + 2 + 3 + \dots + 20)^2 = (1^2 + 2^2 + 3^2 + \dots + 20^2) + 2[1 \cdot 2 + 1 \cdot 3 + \dots + 1 \cdot 20 + 2 \cdot 3 + \dots + 19 \cdot 20]$$

$$\left(\frac{20 \times 21}{2} \right)^2 = \left(\frac{20 \times 21 \times 41}{6} \right) + 2[\alpha]$$

$$a = b + 2\alpha$$

$$\alpha = \frac{a - b}{2}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 23 to 25

Let A_1, G_1, H_1 denote the arithmetic, geometric and harmonic means, respectively, of two distinct positive numbers. For $n \geq 2$, let A_{n-1} and H_{n-1} have arithmetic, geometric and harmonic means as A_n, G_n, H_n respectively.

माना A_1, G_1, H_1 क्रमशः दो भिन्न धनात्मक संख्याओं के समान्तर, गुणोत्तर एवं हरात्मक माध्य है। $n \geq 2$ के लिये माना A_{n-1} एवं H_{n-1} के समान्तर, गुणोत्तर एवं हरात्मक माध्य क्रमशः A_n, G_n, H_n है।

23. Which one of the following statements is correct?

- (A) $G_1 > G_2 > G_3 > \dots$
- (B) $G_1 < G_2 < G_3 < \dots$
- (C) $G_1 = G_2 = G_3 = \dots$
- (D) $G_1 < G_3 < G_5 < \dots$ and $G_2 > G_4 > G_6 > \dots$

निम्नलिखित में से कौनसा कथन सत्य है ?

- (A) $G_1 > G_2 > G_3 > \dots$
- (B) $G_1 < G_2 < G_3 < \dots$
- (C) $G_1 = G_2 = G_3 = \dots$
- (D) $G_1 < G_3 < G_5 < \dots$ तथा $G_2 > G_4 > G_6 > \dots$

Ans. C

Sol. $A_n = \frac{A_{n-1} + H_{n-1}}{2} \quad \dots(1)$

$$G_n = \sqrt{(A_{n-1})(H_{n-1})} \quad \dots(2)$$

$$H_n = \frac{2A_{n-1} \cdot H_{n-1}}{A_{n-1} + H_{n-1}} \quad \dots(3)$$

$$G_n^2 = (A_n)(H_n) \quad \dots(4)$$

By equation (2)

$$G_n^2 = A_{n-1}H_{n-1} = A_nH_n$$

$\Rightarrow G_n$ is independent from n

$$\Rightarrow G_1 = G_2 = G_3 = \dots$$

24. Which one of the following statements is correct?

(A) $A_1 > A_2 > A_3 > \dots$

(B) $A_1 < A_2 < A_3 < \dots$

(C) $A_1 > A_3 > A_5 > \dots$ and $A_2 < A_4 < A_6 < \dots$

(D) $A_1 < A_3 < A_5 < \dots$ and $A_2 > A_4 > A_6 > \dots$

निम्नलिखित में से कौनसा कथन सत्य है ?

(A) $A_1 > A_2 > A_3 > \dots$

(B) $A_1 < A_2 < A_3 < \dots$

(C) $A_1 > A_3 > A_5 > \dots$ तथा $A_2 < A_4 < A_6 < \dots$

(D) $A_1 < A_3 < A_5 < \dots$ तथा $A_2 > A_4 > A_6 > \dots$

Ans. A

Sol. By (1)

$$A_n = \frac{A_{n-1} + H_{n-1}}{2}$$

$$\because A_{n-1} \geq H_{n-1}$$

$$\because \text{numbers are distinct, so } A_{n-1} > H_{n-1}$$

$$\Rightarrow A_n < \frac{A_{n-1} + A_{n-1}}{2}$$

$$A_n < A_{n-1}$$

$$A_1 > A_2 > A_3 > \dots$$

25. Which one of the following statements is correct?

(A) $H_1 > H_2 > H_3 > \dots$

(B) $H_1 < H_2 < H_3 < \dots$

(C) $H_1 > H_3 > H_5 > \dots$ and $H_2 < H_4 < H_6 < \dots$

(D) $H_1 < H_3 < H_5 < \dots$ and $H_2 > H_4 > H_6 > \dots$

निम्नलिखित में से कौनसा कथन सत्य है ?

(A) $H_1 > H_2 > H_3 > \dots$

(B) $H_1 < H_2 < H_3 < \dots$

(C) $H_1 > H_3 > H_5 > \dots$ तथा $H_2 < H_4 < H_6 < \dots$

(D) $H_1 < H_3 < H_5 < \dots$ तथा $H_2 > H_4 > H_6 > \dots$

Ans. B

Sol.

$$H_n = \frac{2}{\frac{1}{A_{n-1}} + \frac{1}{H_{n-1}}} \quad \Rightarrow A_n < \frac{A_{n-1} + A_{n-1}}{2}$$

$$\frac{1}{A_{n-1}} + \frac{1}{H_{n-1}} = \frac{2}{H_n}$$

$$\because A_{n-1} > H_{n-1}$$

$$\frac{1}{H_{n-1}} > \frac{1}{A_{n-1}}$$

$$\frac{2}{H_{n-1}} > \frac{1}{A_{n-1}} + \frac{1}{H_{n-1}}$$

$$\frac{2}{H_{n-1}} > \frac{2}{H_n}$$

$$H_n > A_{n-1}$$

$$H_1 < H_2 < H_3 < \dots$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

26. In an A.P., let T_r denote r^{th} term from beginning, $T_p = \frac{1}{q(p+q)}$, $T_q = \frac{1}{p(p+q)}$, then :

(A) $T_1 = \text{common difference}$

(B) $T_{p+q} = \frac{1}{pq}$

(C) $T_{pq} = \frac{1}{p+q}$

(D) $T_{p+q} = \frac{1}{p^2q^2}$

समान्तर श्रेणी में माना T_r , $r^{\text{वें}}$ पद को निरूपित करता है (प्रारम्भ से), $T_p = \frac{1}{q(p+q)}$, $T_q = \frac{1}{p(p+q)}$, तो :

(A) $T_1 = \text{सार्वअन्तर}$

(B) $T_{p+q} = \frac{1}{pq}$

(C) $T_{pq} = \frac{1}{p+q}$

(D) $T_{p+q} = \frac{1}{p^2q^2}$

Ans. ABC

Sol. $T_r = a + (r-1)d$

$$T_p = a + (p-1)d = \frac{1}{q(p+q)} \quad \dots(1)$$

$$T_q = a + (q-1)d = \frac{1}{p(p+q)} \quad \dots(2)$$

Solving (1), (2)

$$a = d = \frac{1}{pq(p+q)}$$

T_1 = Common difference

$$T_{p+q} = a + (p+q-1)d$$

$$= (p+q)d$$

$$= (p+q) \left(\frac{1}{pq(p+q)} \right)$$

$$T_{p+q} = \frac{1}{pq}$$

$$T_{pq} = a + (pq-1)d$$

$$= pq \left(\frac{1}{pq(p+q)} \right)$$

$$T_{pq} = \frac{1}{p+q}$$

27. All roots of equation $x^5 - 40x^4 + \alpha x^3 + \beta x^2 + \gamma x + \delta = 0$ are in G.P. If the sum of their reciprocals is 10, then δ can be equal to :

समीकरण $x^5 - 40x^4 + \alpha x^3 + \beta x^2 + \gamma x + \delta = 0$ के सभी मूल गुणोत्तर श्रेणी में हैं। यदि उनके व्युत्क्रमों का योग 10 है, तो δ का मान हो सकता है :

- (A) 32 (B) -32 (C) $\frac{1}{32}$ (D) $-\frac{1}{32}$

Ans. AB

Sol. $x^5 - 40x^4 + \alpha x^3 + \beta x^2 + \gamma x + \delta = 0$

Let $\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$ are roots of polynomial

$$\left[\frac{a}{r^2} + \frac{a}{r} + a + ar + ar^2 \right] = 40$$

$$\left(\frac{1}{r^2} + r^2 \right) + \left(\frac{1}{r} + r \right) + 1 = \frac{40}{a} \quad \dots(1)$$

$$\frac{r^2}{a} + \frac{r}{a} + \frac{1}{a} + \frac{1}{ar} + \frac{1}{ar^2} = 10$$

$$\left(\frac{1}{r^2} + r^2\right) + \left(r + \frac{1}{r}\right) + 1 = 10a \quad \dots(2)$$

Solving (1), (2)

$$10a = \frac{40}{a}$$

$$a^2 = 4$$

$$a = \pm 2$$

$$\delta = -\left(\frac{a}{r^2} \times \frac{a}{r} \times a \times ar \times ar^2\right)$$

$$\delta = -a^5$$

$$\text{at } a = 2$$

$$\delta = -32$$

$$a = -2$$

$$\delta = 32$$

28. For the A.P. given by $a_1, a_2, \dots, a_n, \dots$, the equations satisfied are :

समान्तर श्रेणी $a_1, a_2, \dots, a_n, \dots$ के लिए निम्न में से कौनसा सम्बन्ध सत्य है :

$$(A) a_1 + 2a_2 + a_3 = 0$$

$$(B) a_1 - 2a_2 + a_3 = 0$$

$$(C) a_1 + 3a_2 - 3a_3 - a_4 = 0$$

$$(D) a_1 - 4a_2 + 6a_3 - 4a_4 + a_5 = 0$$

Ans. BD

Sol. $a_1, a_2, a_3, \dots, a_n, \dots$ are in A.P.

$$(A) 2a_2 = a_1 + a_3$$

$$\Rightarrow a_1 + 2a_2 + a_3 \neq 0$$

$$(B) a_1 - 2a_2 + a_3 = 0$$

$$(C) a_1 + 3a_2 - 3a_3 - a_4 = a_1 + 3(a_1 + d) - (a_1 + 2d) - (a_1 + 3d) \neq 0$$

$$(D) a_1 - 4a_2 + 6a_3 - 4a_4 + a_5$$

$$= a_1 - 4(a_1 + d) + 6(a_1 + 2d) - 4(a_1 + 3d) + (a_1 + 4d) = 0$$

29. If sum of the infinite G.P., $p, 1, \frac{1}{p}, \frac{1}{p^2}, \frac{1}{p^3}, \dots$ is $\frac{9}{2}$, the value of p is :

यदि अनन्त गु.श्रे. $p, 1, \frac{1}{p}, \frac{1}{p^2}, \frac{1}{p^3}, \dots$ का योगफल $\frac{9}{2}$ हो, तो p का मान है :

$$(A) 3$$

$$(B) \frac{2}{3}$$

$$(C) \frac{3}{2}$$

$$(D) \frac{1}{3}$$

Ans. AC

Sol. $\frac{p}{1 - \frac{1}{p}} = \frac{9}{2}$

$$\frac{p^2}{p-1} = \frac{9}{2}$$

$$2p^2 = 9p - 9$$

$$2p^2 - 9p + 9 = 0$$

$$2p^2 - 6p - 3p + p = 0$$

$$2p(p-3) - 3(p-3) = 0$$

$$p = \frac{3}{2}, p = 3$$

30. If the arithmetic mean of two positive numbers a & b ($a > b$) is twice their geometric mean, then $a : b$ is :

यदि दो धनात्मक संख्याओं a एवं b ($a > b$) का समान्तर माध्य उनके गुणोत्तर माध्य का दुगुना हो, तो $a : b$ है :

(A) $2 + \sqrt{3} : 2 - \sqrt{3}$ (B) $7 + 4\sqrt{3} : 1$ (C) $1 : 7 - 4\sqrt{3}$ (D) $2 : \sqrt{3}$

Ans. ABC

Sol. $\frac{a+b}{2} = 2\sqrt{ab}$

$$(a+b) = 4\sqrt{ab}$$

$$(a+b)^2 = 16ab$$

$$a^2 + b^2 - 14ab = 0$$

$$a^2 - 14ab + b^2 = 0$$

$$a = \frac{14b \pm \sqrt{(-14b)^2 - 4 \times b^2}}{2}$$

$$a = \frac{14b \pm \sqrt{192b^2}}{2}$$

$$a = \frac{14b \pm 8\sqrt{3}b}{2}$$

$$\frac{a}{b} = 7 \pm 4\sqrt{3}$$

$$\frac{a}{b} = \frac{7+4\sqrt{3}}{1} \quad \text{or} \quad \frac{a}{b} = \frac{7-4\sqrt{3}}{1}$$

$$\frac{a}{b} = \frac{7+4\sqrt{3}}{1} \times \frac{7-4\sqrt{3}}{7-4\sqrt{3}} = \frac{1}{7-4\sqrt{3}}$$

$$\frac{a}{b} = \frac{2+\sqrt{3}}{2\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = \frac{7+4\sqrt{3}}{1}$$

31. If $\sum_{r=1}^n r(r+1)(2r+3) = an^4 + bn^3 + cn^2 + dn + e$, then :

(A) $a + c = b + d$

(B) $e = 0$

(C) $a, b - 2/3, c - 1$ are in A.P.

(D) c/a is an integer

यदि $\sum_{r=1}^n r(r+1)(2r+3) = an^4 + bn^3 + cn^2 + dn + e$ हो, तो :

(A) $a + c = b + d$

(B) $e = 0$

(C) $a, b - 2/3, c - 1$ स.श्रे. में है

(D) c/a एक पूर्णांक है

Ans. ABCD

Sol. $\sum_{r=1}^n (r+1)(2r+1) = 2\sum_{r=1}^n r^3 + 5\sum_{r=1}^n r^2 + 3\sum_{r=1}^n r$

$$an^4 + bn^3 + cn^2 + dn + e = 2\left[\frac{n(n+1)}{2}\right]^2 + 5\left[\frac{n(n+1)(2n+1)}{6}\right] + 3\left[\frac{n(n+1)}{2}\right]$$

$$= \frac{n(n+1)}{2} \left[n(n+1) + \frac{5(2n+1)}{3} + 3 \right]$$

$$= \left(\frac{n^2 + n}{2} \right) \left[\frac{3n^2 + 3n + 10n + 5 + 9}{3} \right]$$

$$= \left(\frac{n^2 + n}{6} \right) (3n^2 + 13n + 14)$$

$$an^4 + bn^3 + cn^2 + dn + e = \frac{1}{6} (3n^4 + 16n^3 + 27n^2 + 14n)$$

$$e = 0, d = \frac{14}{6}, c = \frac{27}{6}, b = \frac{16}{6}, a = \frac{3}{6}$$

$$a + c = \frac{3}{6} + \frac{27}{6} = 5 = b + d$$

(C) $a, b - \frac{2}{3}, c - 1 \Rightarrow \frac{3}{6}, \frac{16}{6} - \frac{2}{3}, \frac{27}{6} - 1$

$$\Rightarrow \frac{1}{2}, 2, \frac{7}{2}$$

$$2 \times 2 = \frac{1}{2} + \frac{7}{2} = 4$$

a, $b - \frac{2}{3}$, c - 1 are in A.P.

$$(D) \frac{c}{a} = \frac{\left(\frac{27}{6}\right)}{\left(\frac{3}{6}\right)} = 9$$

32. If the first two terms of a progression are 8 and 4 respectively, then correct statement(s) is/are:

(A) if third term is 2, then the terms are in G. P.

(B) if third term is 0, then the terms are in A. P.

(C) if third term is $\frac{8}{3}$, then the terms are in H. P.

(D) if the third term is 3, then terms are in A. P.

यदि अनुक्रम के पहले दो पद क्रमशः 8 तथा 4 हैं, तो कौनसा/कौनसे कथन सत्य होंगे :

(A) यदि तीसरा पद 2 है, तो पद गुणोत्तर श्रेणी में होंगे

(B) यदि तीसरा पद 0 है, तो पद समान्तर श्रेणी में होंगे

(C) यदि तीसरा पद $\frac{8}{3}$ है, तो पद हरात्मक श्रेणी में होंगे

(D) यदि तीसरा पद 3 है, तो पद समान्तर श्रेणी में होंगे

Ans. ABC

Sol. $T_1 = 8, T_2 = 4$

(A) $T_3 = 2$

8, 4, 2,

$$\frac{T_3}{T_2} = \frac{2}{4} = \frac{1}{2} = \frac{T_2}{T_1}$$

$\Rightarrow$ Geometric progression

(B) $8, 4, \frac{8}{3}, \dots$

$$\frac{2 \times (8) \left(\frac{8}{3}\right)}{8 + \frac{8}{3}} = \frac{128}{32} = 4$$

$\Rightarrow$ H.P.

(B) 8, 4, 0,

$$2 \times 4 = 8 + 0$$

$\Rightarrow$ A.P.

(D) 8, 4, 3,

$$2 \times 4 \neq 8 + 3$$

Not in A.P.

33. If $40 + \frac{155}{4} + \frac{150}{4} + \frac{145}{4} + \dots + n$ terms = 625 then n can be :

- (A) 24 (B) 25 (C) 40 (D) 37

यदि $40 + \frac{155}{4} + \frac{150}{4} + \frac{145}{4} + \dots + n$ पद = 625 है, तो n का मान हो सकता है :

- (A) 24 (B) 25 (C) 40 (D) 37

Ans. BC

Sol. $40 + \frac{155}{4} + \frac{100}{4} + \frac{145}{4} + \dots + n$ terms = 625

$$160 + 155 + 150 + \dots + \dots + n \text{ terms} = (625) \times 4$$

$$\frac{n}{2} [2 \times 160 + (n-1)(-5)] = 625 \times 4$$

$$\frac{n}{2} [325 - 5n] = 2500$$

$$325n - 5n^2 = 5000$$

$$5n^2 - 325n + 5000 = 0$$

$$n^2 - 65n + 1000 = 0$$

$$n = \frac{65 \pm 15}{2}$$

$$n = 40, n = 25$$

MATCH THE COLUMN TYPE QUESTION(S)

34. Match the following columns :

Column – I

Column – II

(A) If $\log_5 2$, $\log_5(2^x - 5)$ and $\log_5(2^x - 7/2)$ are in A.P., then value of $2x$ is equal to (P) 6

(B) Let S_n denote sum of first n terms of an A.P. If $S_{2n} = 3S_n$, then $\frac{S_{3n}}{S_n}$ is (Q) 9

(C) Sum of infinite series $4 + \frac{8}{3} + \frac{12}{3^2} + \frac{16}{3^3} + \dots$ is (R) 3

- (D) The length, breadth, height of a rectangular box are in G.P. The volume is 27, the total surface area is 78. Then the length is (S) 1

निम्न कॉलम को सुमेलित कीजिए :

Column – I

Column – II

- (A) यदि $\log_5 2, \log_5(2^x - 5)$ तथा $\log_5(2^x - 7/2)$ स. श्रे. में हैं, तो $2x$ का मान होगा (P) 6
 (B) माना S_n एक स. श्रे. के प्रथम n पदों को निरूपित करता है यदि $S_{2n} = 3S_n$ तो, $\frac{S_{3n}}{S_n}$ का मान होगा (Q) 9

- (C) अनन्त श्रेणी $4 + \frac{8}{3} + \frac{12}{3^2} + \frac{16}{3^3} + \dots$ का योग होगा (R) 3
 (D) एक आयताकार बक्से के लम्बाई, चौड़ाई तथा ऊँचाई गु. श्रे. में हैं इसका आयतन 27 है, कुल पृष्ठीय क्षेत्रफल 78 है, तो लम्बाई होगी (S) 1

Ans. (A) P, (B) P, (C) q, (D) Q, S

Sol. (A) $\log_5 2, \log_5(2^x - 5), \log_5\left(2^x - \frac{7}{2}\right)$ are in A.P.

$$2\log_5(2^x - 5) = \log_5 + 2\log_5\left(2^x - \frac{7}{2}\right)$$

$$(2^x - 5)^2 = 2\left(2^x - \frac{7}{2}\right)$$

Let $t = 2^x$

$$(t - 5)^2 = 2\left(t - \frac{7}{2}\right)$$

$$t^2 + 25 - 10t = 2t - 7$$

$$t^2 - 12t + 32 = 0$$

$$(t - 8)(t - 4) = 0$$

$$t = 4, 8$$

$$2^x = 2 \quad \text{or} \quad 2^x = 8$$

$$x = 1 \quad \text{or} \quad x = 3$$

$$2x = 2 \quad \text{or} \quad 2x = 6$$

$$\text{But } 2^x - 5 > 0$$

$$2^x > 5$$

$$\Rightarrow x > 2 \quad \Rightarrow \quad x = 3$$

$$(B) S_n = \frac{n}{2}[2a + (n-1)d]$$

$$S_{2n} = 3S_n$$

$$\frac{2n}{2}[2a + (2n-1)d] = 3 \times \frac{n}{2}[2a + (n-1)d]$$

$$2[2a + (2n-1)d] = 3[2a + (n-1)d]$$

$$4a + (4n-2)d = 6a + (3n-3)d$$

$$2a = (n+1)d$$

$$\frac{S_{3n}}{S_n} = \frac{\frac{3n}{2}[2a + (3n-1)d]}{\frac{n}{2}[2a + (n-1)d]} = \frac{3[(n+1) + (3n-1)]}{[(n+1) + (n-1)]}$$

$$= 3 \left[\frac{4n}{2n} \right]$$

$$= 6$$

$$(C) S = 4 + \frac{8}{3} + \frac{12}{3^2} + \frac{16}{3^3} + \dots \dots \dots \quad \dots(1)$$

$$\frac{S}{3} = \frac{4}{3} + \frac{8}{3^2} + \frac{12}{3^3} + \dots \dots \dots \quad \dots(2)$$

$$(1) - (2)$$

$$\frac{2S}{3} = 4 + \frac{4}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \dots \dots \dots$$

$$\frac{2S}{3} = 4 \left[\frac{1}{1 - \frac{1}{3}} \right]$$

$$S = \frac{3}{2} \times 4 \left(\frac{3}{2} \right) = 9$$

(D) let length = a

breadth = b,

Height = c

a, b, c are in G.P. $\Rightarrow \frac{a}{r}, a, ar$

volume of rectangular box = $abc = 27$

$$\left(\frac{a}{r} \right) (a) (ar) = 27$$

$$a^3 = 3^3$$

$$a = 3$$

Surface area = $2(ab + bc + ca) = 78$

$$2 \left[\left(\frac{a}{r} \right) (a) + a(ar) + (ar) \left(\frac{a}{r} \right) \right] = 78$$

$$a^2 \left[\frac{r^2 + r + 1}{r} \right] = 39$$

$$9 \left[\frac{r^2 + r + 1}{r} \right] = 39$$

$$3r^2 + 3r + 3 = 13r$$

$$3r^2 - 10r + 3 = 0$$

$$3r^2 - 9r - r + 3 = 0$$

$$3r(r - 3) - (r - 3) = 0$$

$$(3r - 1)(r - 3) = 0$$

$$r = 3, \frac{1}{3}$$

$$\text{length} = \frac{a}{r} = \frac{3}{3} \text{ or } \left(\frac{3}{\frac{1}{3}} \right)$$

$$= 1 \text{ or } 9$$

35. Match the following columns :

Column – I

(A) If $\log_x y, \log_z x, \log_y z$ are in G.P., $xyz = 64$ and x^3, y^3, z^3 are in A.P.,

$$\text{then } \frac{3x}{y} =$$

(B) The value of $2^{\frac{1}{4}} \cdot 4^{\frac{1}{8}} \cdot 8^{\frac{1}{16}} \dots \infty =$

(C) If x, y, z are in A.P., then $(x + 2y - z)(2y + z - x)(z + x - y) = kxyz$,

where $k \in \mathbb{N}$, then $k =$

Column – II

(P) 2

(Q) 1

(R) 3

(S) 4

निम्न कॉलम को सुमेलित कीजिए :

Column – I

(A) यदि $\log_x y, \log_z x, \log_y z$ गु.श्रे. में हो, $xyz = 64$ तथा x^3, y^3, z^3 स.श्रे. में हो तो

$$\frac{3x}{y} =$$

(B) $2^{\frac{1}{4}} \cdot 4^{\frac{1}{8}} \cdot 8^{\frac{1}{16}} \dots \infty =$

(C) यदि x, y, z स.श्रे. में हो तो $(x + 2y - z)(2y + z - x)(z + x - y) = kxyz$,

जहाँ $k \in \mathbb{N}$, तो $k =$

Column – II

(P) 2

(Q) 1

(R) 3

(S) 4

Ans. (A) R, (B) P, (C) S,

Sol. $\log_x y, \log_z x, \log_y z$ are in G.P.

$$(\log_z x)^2 = (\log_x y)(\log_y z)$$

$$(\log_z x)^2 = (\log_x z)$$

$$(\log_z x)^3 = 1$$

$$\log_z x = 1 \Rightarrow x = z$$

$$xyz = 64$$

$$x^2 y = 64 \Rightarrow y = \frac{64}{x^2}$$

x^3, y^3, z^3 are in A.P.

$$2y^3 = x^3 + z^3$$

$$y^3 = x^3$$

$$\left(\frac{64}{x^2}\right)^3 = x^3$$

$$(64)^3 = x^3 \cdot x^6$$

$$4^9 = x^9$$

$$x = 4 \Rightarrow y = 4 \Rightarrow z = 4$$

$$\frac{3x}{y} = 3$$

$$(B) S = 2^{\frac{1}{4}} \cdot 4^{\frac{1}{8}} \cdot 8^{\frac{1}{16}} \dots = 2^{\frac{1}{4}} \cdot 2^{\frac{2}{8}} \cdot 2^{\frac{3}{16}}$$

$$S = 2^{\left(\frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \dots\right)}$$

$$A = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \dots \quad \dots(1)$$

$$\frac{A}{2} = 0 + \frac{1}{8} + \frac{2}{16} + \dots \quad \dots(2)$$

$$(1) - (2)$$

$$\frac{A}{2} = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

$$\frac{A}{2} = \frac{1}{4} \left[1 + \frac{1}{2} + \frac{1}{4} + \dots \right] = \frac{1}{4} \left[\frac{1}{1 - \frac{1}{2}} \right] = \frac{1}{2}$$

$$A = 1$$

$$S = 2^1 = 2$$

$$(C) (x + 2y - z)(2y + z - x)(z + x - y) = k(xyz)$$

x, y, z are in A.P.

$$2y = x + z$$

$$(x + x + z - z)(x + z + z - x) \left(z + x - \left(\frac{x+z}{2} \right) \right) = k(xyz)$$

$$(2x)(2z) \left(\frac{x+z}{2} \right) = k(xyz)$$

$$4(xyz) = k(xyz)$$

$$k = 4$$

36. Match the series in Column I with their sum in Column II :

Column I

(A) $\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots$ to infinity equals

(B) $\frac{2^3-1}{2^3+1} \times \frac{3^3-1}{3^3+1} \times \frac{4^3-1}{4^3+1} \dots$ to infinity equals

(C) $\frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \dots$ to infinity equals

(D) $\frac{1}{1.2.3.4} + \frac{1}{2.3.4.5} + \frac{1}{3.4.5.6} + \dots$ to infinity equals

Column II

(P) $\frac{1}{4}$

(Q) $\frac{1}{2}$

(R) $\frac{2}{3}$

(S) $\frac{1}{18}$

कॉलम I में दी गई श्रेणी का कॉलम II में दिये गये योग से मिलान कीजिए :

Column I

(A) $\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots$ अनन्त तक

(B) $\frac{2^3-1}{2^3+1} \times \frac{3^3-1}{3^3+1} \times \frac{4^3-1}{4^3+1} \dots$ अनन्त तक

(C) $\frac{1}{1.2.3} + \frac{1}{2.3.4} + \frac{1}{3.4.5} + \dots$ अनन्त तक

(D) $\frac{1}{1.2.3.4} + \frac{1}{2.3.4.5} + \frac{1}{3.4.5.6} + \dots$ अनन्त तक

Column II

(P) $\frac{1}{4}$

(Q) $\frac{1}{2}$

(R) $\frac{2}{3}$

(S) $\frac{1}{18}$

Ans. (A) Q, (B) R, (C) P, (D) S

Sol. (A) $S = \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots$

$$T_r = \left[1 - \frac{1}{(r+1)^2} \right] = \frac{(r+1)^2 - 1}{(r+1)^2} = \frac{r(r+2)}{(r+1)^2}$$

$$S = \prod_{r=1}^{\infty} T_r = \left(\frac{1 \cdot 3}{2^2} \right) \left(\frac{2 \cdot 4}{3^2} \right) \left(\frac{3 \cdot 5}{4^2} \right) \left(\frac{4 \cdot 6}{5^2} \right) \dots$$

$$= \frac{1}{2}$$

$$(B) S = \frac{2^3 - 1}{2^3 + 1} \times \frac{3^3 - 1}{3^3 + 1} \times \frac{4^3 - 1}{4^3 + 1} \times \dots$$

$$T_r = \frac{(r+1)^3 - 1}{(r+1)^3 + 1} = \frac{((r+1) - 1)((r+1) + 1)}{((r+1) + 1)} \times \left[\frac{(r+1)^2 + 1 + (r+1)}{(r+1)^2 + 1 - (r+1)} \right]$$

$$= \left(\frac{r}{r+2} \right) \left[\frac{r^2 + 3r + 3}{r^2 + r + 1} \right]$$

$$S = \prod_{r=1}^{\infty} T_r$$

$$S = \prod_{r=1}^{\infty} \left(\frac{r}{r+2} \right) \left(\frac{r^2 + 3r + 3}{r^2 + r + 1} \right)$$

$$= \left(\frac{1}{3} \right) \left(\frac{7}{3} \right) \times \left(\frac{2}{4} \right) \left(\frac{13}{7} \right) \times \left(\frac{3}{5} \right) \left(\frac{21}{13} \right) \times \dots$$

$$S = \frac{2}{3}$$

$$(C) S = \frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \dots$$

$$S = \sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} = \sum_{n=1}^{\infty} \left[\frac{(n+2) - n}{n(n+1)(n+2)} \right]$$

$$S = \frac{1}{2} \sum_{n=1}^{\infty} \left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right]$$

$$S = \frac{1}{2} \left[\left(\frac{1}{1 \cdot 2} - \frac{1}{2 \cdot 3} \right) + \left(\frac{1}{2 \cdot 3} - \frac{1}{3 \cdot 4} \right) + \dots \right]$$

$$S = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$(D) S = \frac{1}{1 \cdot 2 \cdot 3 \cdot 4} + \frac{1}{2 \cdot 3 \cdot 4 \cdot 5} + \dots$$

$$\begin{aligned}
S &= \sum_{r=1}^{\infty} \frac{1}{n(n+1)(n+2)(n+3)} = \frac{1}{3} \sum_{n=1}^{\infty} \frac{(n+3) - n}{n(n+1)(n+2)(n+3)} \\
S &= \frac{1}{3} \sum_{n=1}^{\infty} \left[\frac{1}{n(n+1)(n+2)} - \frac{1}{(n+1)(n+2)(n+3)} \right] \\
S &= \frac{1}{6} \sum_{n=1}^{\infty} \left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right] - \frac{1}{6} \sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)} - \frac{1}{(n+2)(n+3)} \\
&= \frac{1}{6} \sum_{n=1}^{\infty} \frac{1}{n(n+1)} - \frac{1}{3} \sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)} + \frac{1}{6} \sum_{n=1}^{\infty} \frac{1}{(n+2)(n+3)} \\
&= \frac{1}{6} \left[\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) \right] - \frac{1}{3} \left[\sum_{n=1}^{\infty} \left(\frac{1}{n+1} - \frac{1}{n+2} \right) \right] + \frac{1}{6} \left[\sum_{n=1}^{\infty} \left(\frac{1}{n+2} - \frac{1}{n+3} \right) \right] \\
&= \frac{1}{6} - \frac{1}{6} + \frac{1}{6} \times \frac{1}{3} \\
&= \frac{1}{18}
\end{aligned}$$